Sequences & Series

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Arithmetic series**

For the model $f(x) = 2x + 4$ in arithmetic series, evaluate $f(5)$.

Final Answer: 14. Substitute $x = 5$: $f(5) = 2(5) + 4 = 14$.

2 **[Foundational]** • **Target: Arithmetic series**

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

Final Answer: $x = 4$ and $x = 8$. Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

3 **[Foundational]** • **Target: Geometric sequences**

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 104°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 45^\circ) = 104^\circ$.

4 **[Foundational]** • **Target: Geometric sequences**

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 10 cm. Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

5 **[Foundational]** • **Target: Geometric sequences**

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 10. Substituting $x = 1$ into $y = 5x + 5$ gives $y = 5(1) + 5 = 10$.

6 **[Foundational]** • **Target: Geometric sequences**

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 97°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 52^\circ) = 97^\circ$.

7 **[Foundational]** • **Target: Geometric sequences**

A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 15 cm. Multiplying the original length by the scale factor gives $5 \times 3 = 15$ cm.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Geometric sequences**

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 15. Substituting $x = 2$ into $y = 5x + 5$ gives $y = 5(2) + 5 = 15$.

9 **[Intermediate]** • **Target: Geometric sequences**

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 90°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 59^\circ) = 90^\circ$.

10 **[Intermediate]** • **Target: Geometric sequences**

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 20 cm. Multiplying the original length by the scale factor gives $5 \times 4 = 20$ cm.

11 **[Intermediate]** • **Target: Geometric sequences**

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 20. Substituting $x = 3$ into $y = 5x + 5$ gives $y = 5(3) + 5 = 20$.

12 **[Intermediate]** • **Target: Geometric sequences**

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 83°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 66^\circ) = 83^\circ$.

13 **[Intermediate]** • **Target: Geometric sequences**

A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 25 cm. Multiplying the original length by the scale factor gives $5 \times 5 = 25$ cm.

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14 [Intermediate] • Target: Geometric sequences

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Final Answer: 25. Substituting $x = 4$ into $y = 5x + 5$ gives $y = 5(4) + 5 = 25$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Geometric series • Cross-Lesson: 11.01 Probability & Statistics

In ΔABC studying geometric series, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 103°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 45^\circ) = 103^\circ$.

16 [Challenging] • Target: Geometric series • Cross-Lesson: 2.01 Quadratics & Complex Numbers

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 12 cm. Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

17 [Challenging] • Target: Geometric series • Cross-Lesson: 3.01 Polynomial Functions

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 8. Substituting $x = 1$ into $y = 2x + 6$ gives $y = 2(1) + 6 = 8$.

18 [Challenging] • Target: Geometric series • Cross-Lesson: 4.01 Rational Expressions

In ΔABC studying geometric series, $m\angle A = 32^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 96°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 52^\circ) = 96^\circ$.

19 [Challenging] • Target: Geometric series • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 18 cm. Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

20 [Challenging] • Target: Geometric series • Cross-Lesson: 6.01 Radicals & Rational Exponents

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 10. Substituting $x = 2$ into $y = 2x + 6$ gives $y = 2(2) + 6 = 10$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Notation for sequences and series

For the model $f(x) = 5x + 9$ in notation for sequences and series, evaluate $f(2)$.

a. 19 ✓

b. 24

c. 10

d. 23

Substitute $x = 2$: $f(2) = 5(2) + 9 = 19$.

22 [Foundational] • Target: Notation for sequences and series

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

23 [Foundational] • Target: Notation for sequences and series

For the model $f(x) = 5x + 9$ in notation for sequences and series, evaluate $f(5)$.

a. 34 ✓

b. 39

c. 25

d. 38

Substitute $x = 5$: $f(5) = 5(5) + 9 = 34$.

24 [Foundational] • Target: Arithmetics sequences

For the model $f(x) = 6x + 3$ in arithmetics sequences, evaluate $f(2)$.

a. 15 ✓

b. 21

c. 12

d. 19

Substitute $x = 2$: $f(2) = 6(2) + 3 = 15$.

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25 [Foundational] • Target: Arithmetics sequences

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

26 [Foundational] • Target: Arithmetics sequences

For the model $f(x) = 6x + 3$ in arithmetics sequences, evaluate $f(5)$.

a. 33 ✓

b. 39

c. 30

d. 37

Substitute $x = 5$: $f(5) = 6(5) + 3 = 33$.

27 [Foundational] • Target: Arithmetic series

For the model $f(x) = 2x + 4$ in arithmetic series, evaluate $f(2)$.

a. 8 ✓

b. 10

c. 4

d. 12

Substitute $x = 2$: $f(2) = 2(2) + 4 = 8$.

28 [Foundational] • Target: Arithmetic series

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Notation for sequences and series

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

30 [Intermediate] • Target: Notation for sequences and series

A data set recorded for notation for sequences and series consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

31 [Intermediate] • Target: Notation for sequences and series

For the model $f(x) = 5x + 9$ in notation for sequences and series, evaluate $f(4)$.

a. 29 ✓

b. 34

c. 20

d. 33

Substitute $x = 4$: $f(4) = 5(4) + 9 = 29$.

32 [Intermediate] • Target: Notation for sequences and series

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

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33 [Intermediate] • Target: Arithmetics sequences

A data set recorded for arithmetics sequences consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

34 [Intermediate] • Target: Arithmetics sequences

For the model $f(x) = 6x + 3$ in arithmetics sequences, evaluate $f(4)$.

a. 27 ✓

b. 33

c. 24

d. 31

Substitute $x = 4$: $f(4) = 6(4) + 3 = 27$.

35 [Intermediate] • Target: Arithmetics sequences

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

36 [Intermediate] • Target: Arithmetic series

A data set recorded for arithmetic series consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

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37 [Intermediate] • Target: Arithmetic series

For the model $f(x) = 2x + 4$ in arithmetic series, evaluate $f(4)$.

a. 12 ✓

b. 14

c. 8

d. 16

Substitute $x = 4$: $f(4) = 2(4) + 4 = 12$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric sequences • Cross-Lesson: 4.01 Rational Expressions

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm ✓

c. 15 cm

d. 20 cm

Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

39 [Challenging] • Target: Geometric sequences • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 97° ✓

b. 107°

c. 89°

d. 149°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 52^\circ) = 97^\circ$.

40 [Challenging] • Target: Geometric sequences • Cross-Lesson: 6.01 Radicals & Rational Exponents

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 90° ✓

b. 100°

c. 82°

d. 149°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 59^\circ) = 90^\circ$.

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41 [Challenging] • Target: Geometric sequences • Cross-Lesson: 8.01 Conic Sections

A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm ✓

c. 30 cm

d. 125 cm

Multiplying the original length by the scale factor gives $5 \times 5 = 25$ cm.

42 [Challenging] • Target: Geometric series • Cross-Lesson: 9.01 Trigonometric Functions

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm ✓

c. 18 cm

d. 24 cm

Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

43 [Challenging] • Target: Geometric series • Cross-Lesson: 1.01 Functions & Their Graphs

In ΔABC studying geometric series, $m\angle A = 32^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 96° ✓

b. 106°

c. 88°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 52^\circ) = 96^\circ$.

44 [Challenging] • Target: Geometric series • Cross-Lesson: 10.01 Matrices & Systems

In ΔABC studying geometric series, $m\angle A = 32^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 89° ✓

b. 99°

c. 81°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 59^\circ) = 89^\circ$.

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45 [Challenging] • Target: Geometric series • Cross-Lesson: 11.01 Probability & Statistics

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm ✓

c. 36 cm

d. 150 cm

Multiplying the original length by the scale factor gives $6 \times 5 = 30$ cm.

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